

Melvin Boateng

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Portfolio: <https://melvin-portfolio-2026.vercel.app/>

EDUCATION

University of the Pacific, Stockton, CA

Bachelor of Science, Computer Science

Master of Science, Computer Science (Accelerated Program)

Minor in Media X

Expected December 2027

GPA: 3.87

Relevant Coursework: Discrete Mathematics, Data Structures, Application Development, Design/Analysis of Algorithms, Computer Game Technologies, Virtual Reality, Secure Software Systems, Computer Systems & Networks, Computing Theory, Data Analytics Programming, Database Management Systems, Storytelling and Visualization, Human/Brain Machine Interface

EXPERIENCE

Climate Change Research Assistant, Stockton, CA

January 2025 – June 2025

- Analyzed 15+ peer-reviewed papers on renewable energy optimization models to evaluate effectiveness and applicability for building an optimal energy portfolio
- Developed MATLAB-based optimization model that minimized carbon emissions by more than 50% on average while meeting electricity demands, demonstrating critical importance of model reliability
- Constructed research paper discussing climate change mitigation through optimization models and government policies, resulting in presentation at departmental symposium
- Evaluated trade-offs between energy sources using quantitative analysis to inform policy recommendations

Science Olympiad Treasurer, Stockton, CA

February 2023 – May 2023

- Co-founded Science Olympiad club at Weston Ranch High School by recruiting 20+ members and establishing organizational structure
- Managed club finances and organized three fundraisers that raised \$500+ for competition fees and materials
- Coordinated with school administration to secure meeting space and competition transportation

PROJECTS

A Hero's Journey (2D Roguelike Platformer)

August 2025 – October 2025

- Engineered 2D action-platformer in Unity using C# to develop responsive player controls, physics-based combat mechanics, and state-machine logic for fluid character animations
- Utilized Unity's Prefab system and Tilemaps to create modular, reusable game objects, streamlining level design process and reducing development time by 30%
- Managed collaborative development through GitHub version control, maintaining clean commit history and resolving merge conflicts for team of 5 developers
- Designed custom pixel-art visual assets and UI elements to support cohesive aesthetic, integrating Media X design principles

Summoner Simulator (ROBLOX Game)

October 2024 – December 2024

- Independently developed complete pet collection game on ROBLOX using Lua, designing all game systems including pet summoning logic, inventory management, and player progression mechanics
- Created 20+ custom 3D pet models in Blender with optimized polygon counts for performance, implementing detailed textures and rigging for animations
- Designed comprehensive UI system from scratch including pet selection menus, inventory displays, and real-time stat tracking interfaces using ROBLOX Studio GUI tools
- Engineered complete game map with custom terrain, structures, and dynamic lighting system to create immersive player environment

Monster Battle (Java Game)

October 2024 – December 2024

- Led team of 4 developers to create Pokemon-inspired 2D game using Java and GitHub for collaboration, completing project on schedule using MindView for Gantt chart planning

- Designed and implemented 15+ interconnected classes (Badges, Battle, Graphics, Map, Monsters, Trainers, Shop) following UML diagrams created in Lucid
- Applied object-oriented programming principles including inheritance, polymorphism, and encapsulation to ensure maintainable, scalable code architecture
- Created custom game assets using Paint.net and AI tools, integrating graphics seamlessly with game logic

Text-Based Banking System (C++)

April 2024

- Developed secure banking application in C++ with Bank, Customer, BankAccount, Savings, and Checking classes, implementing proper data handling and unique identifier management
- Implemented account security features including SSN validation, unique account number generation, and transaction logging to prevent unauthorized access
- Created user-friendly interface that responded to inputs for deposits, withdrawals, transfers, and balance inquiries across multiple account types
- Designed promotional feature that automatically awarded \$100 bonus for opening first Savings Account, demonstrating business logic implementation

A New Adventure (ROBLOX Open-World Foundation)

Summer 2023

- Created team of 6 developers to build foundation for open-world MMORPG on ROBLOX using Lua and Blender for 3D asset creation
- Generated project schedule and organized bi-weekly team meetings with documented agendas and action items, ensuring consistent progress
- Developed prototype featuring furnished player movement, dynamic lighting system, and collection of outdoor 3D models and character designs for future game development

SKILLS

Programming Languages: C++, Java, Lua, C#, MATLAB

Tools & Software: GitHub, Unity, Visual Studio, ROBLOX Studio, Blender, Lucid, MindView, Paint.net, Adobe Firefly, Microsoft Office Suite, Google Workspace

Game Development: 3D Modeling, UI/UX Design, Game Systems Architecture, Lighting Design, Asset Creation

ADDITIONAL EXPERIENCE

Varsity Soccer Player | Key Club Member | Science Olympiad Competitor